

Team Deliverables Document

Oil Field Data Platform | Tigbourne Capital

Team Leads Nalin, Zain	Team Nalin, Zain, Prat, Min	Organization Tigbourne Capital	POC Target By Aug 31, 2026
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1. What We Are Building

The Product

A data platform that lets oil companies, banks, and chemical suppliers screen any oil field in the world, filter by key parameters, and export analysis — delivered via a clean web interface and API. The competitive moat is Tigbourne's 3+ years of accumulated field data, which competitors cannot replicate quickly. Subscribers pay for ongoing access because the data is constantly updated.

MVP Scope — Build This First

- **Core functionality:** filter oil field data by key parameters, then export results in CSV, PDF, or Word (DOCX) format.
- Optional for MVP: data visualisation graphs based on filtered results.
- **CEOR calculations and NPV modelling come after the MVP is complete** — the MVP is the foundation they sit on.
- **Timeline:** MVP is technically achievable in 1–2 weeks. The official stated timeline is 3 months — this is intentional (under-promise, over-deliver). Buddy expects 3 months.

The Competitive Gap

- **The four biggest energy data companies** (S&P Global, Rystad, Enverus, Wood Mackenzie) cover 80,000+ fields but **none combine field-level data with chemical recovery economics**.
- Data is not sold as raw files — it is delivered as abstracted analysis and API keys providing access to formatted, constantly updated data. The subscription model is what makes this a recurring revenue business.

Who Buys It

Buyer	What They Need From Us	Examples
Chemical Companies	Know which fields match their chemicals — instead of approaching operators blind	Dow, BASF, SNF, Hunting PLC, Huntsman
Oilfield Service Companies	Targeting data for their chemical recovery product lines	SLB, Baker Hughes, Halliburton

Buyer	What They Need From Us	Examples
Investment Banks	Independent valuation of a field before and after chemical recovery	UBS (senior advisor connected), Capital One
Oil Operators	Is it better to use chemicals, drill a new well, or sell? With hard numbers.	ExxonMobil, Chevron, EGPC, PanAmerican
PE / Investment Funds	Pre-screened fields ranked by recovery economics	Tigbourne Capital and comparable firms

- **Target customers: Fortune 500 / Fortune 1,000 only.** Sales handled by Buddy's existing team. Tech team's job is to define the ideal customer profile for them.

2. The Data We Sell

Competitive Moat

Tigbourne has 3+ years of accumulated field data from operator outreach and chemical supplier relationships. This data is the moat — subscribers pay for ongoing access because it is constantly updated. No competitor has the combination of field parameters AND chemical economics in one place.

Field-Level Data (per oil field / well)

Category	What We Store	Why It Matters
Production	Barrels of oil per day; total liquid per day; water per day; water cut % (how much of what comes out of the ground is water)	Shows how far the field has declined and how much oil remains
Reservoir Conditions	API gravity (oil thickness — sweet spot 16–40); temperature °C; salt content; viscosity; porosity; permeability	Determines which chemicals work. Above ~130°C most chemicals stop being effective.
Location	Onshore or offshore; depth; operator name; country / region	Offshore costs dramatically more — changes the recovery economics entirely
Cost Data	Lifting + transport + marketing cost; drilling & completion cost; plug & abandonment (P&A) cost estimate — what it costs to safely close a well at end of life. Note: P&A figures currently missing for most fields — sourcing these is a Week 2 priority. Onshore P&A is cheap; offshore P&A is expensive. Most buyers ignore P&A entirely — including it is a key differentiator of our valuation.	Core inputs for every financial calculation. P&A is essential for a complete NPV.

What the Platform Calculates (post-MVP)

- **Break-even price:** The oil price at which a field stops making money — updated live using Brent, WTI, Dubai price feeds
- **CEOR vs. drilling comparison:** What does it cost to apply chemicals vs. drill a new well for the same oil? The core decision every operator needs to make.
- **Net present value (NPV):** How much is this field worth today? Discounts all future cash flows and subtracts costs including P&A.
- **Early vs. late intervention value:** The most important output: how much MORE profit from applying chemicals at 10% water cut vs. waiting until 80%? Roughly 10x more — this platform shows it precisely for each field.
- **Chemical technology cost benchmarks:** Costs for polymer flooding, surfactant, ASP (a mix of three chemicals), foam, and wettability alteration — from Dow, BASF, SNF, Hunting tech data.

3. The Central Insight This Platform Must Prove

△ Most Important — Stressed Repeatedly in Kickoff Meeting

Applying chemical recovery early (when water cut is low, e.g. 10%) produces roughly 10x more oil than applying the same treatment late (when water cut is 80–90%). The cost is the same either way. But most operators wait too long. This platform quantifies that gap precisely — for every field — and makes the case for early intervention with hard numbers.

- **Water cut is the central variable.** Every financial calculation must be runnable at any water cut level.
- **When you intervene matters more than anything else.** The platform must show this visually and with numbers.

4. Team Roles

Person	Role	Responsibilities
Nalin Mehta	Team Lead + Back-End	Overall project delivery; database; data import pipeline; API; financial calculations engine; AWS setup; security; commodity price feed integration
Zain	Co-Lead + Back-End Developer	Supports Nalin on back-end build; commodity price feed research; database support; API work; P&A cost sourcing; pull request review
Prat	Front-End Developer	Create GitHub repo; build the web interface against the API spec Nalin provides; login/session UI; data export (CSV, PDF, DOCX)
Min	Data & Parsing	Send placeholder data to Prat immediately; SPE paper access and parsing tool; map paper data to database fields; ensure data parser handles Excel format changes

- **Quality over speed.** The POC must be solid enough to demo to UBS and Capital One. No rushing a broken product.

5. Immediate Action Items

This Week — Before Any Code Is Written

These tasks unblock everything else. They should all be done within the first 1–2 days of the project starting.

Who	Action	By When
Prat	Create GitHub repo and add all contributors (Nalin, Zain, Min) as collaborators	June 1
All	Link personal GitHub accounts to the Claude team account for automated commits	June 1–2
Min	Send placeholder data to Prat so he can build and test the frontend without waiting for the real dataset	June 2
All	Review and edit this deliverables doc once Nalin uploads it — flag anything that is wrong or missing	June 2–3
Nalin + Zain	Submit infrastructure cost estimate to Buddy before finalising any cloud or hosting decisions	Before next Buddy meeting

6. Technical Implementation — Week-by-Week

Tech Stack

Back end: FastAPI (Python) + PostgreSQL. Database schema managed using a free tool (no additional database costs). Storage: AWS S3. Authentication: secure login tokens (JWT). API documentation: auto-generated from code (OpenAPI). Version control: GitHub with pull request review before any merge. Claude assists with code generation and writes README notes documenting all changes for team continuity. Claude.md file to include test design rules to prevent fake or meaningless tests.

Infrastructure Costs (Buddy Covers All)

AWS server: estimated \$50–100/month depending on traffic. Domain + hosting: approximately \$200. Team to submit a cost estimate to Buddy before finalising any infrastructure decisions.

Month 1 — June 2026: Foundations + MVP

June 1–7 Week 1	• Prat: GitHub repo created and all contributors added; branching strategy agreed (main / staging / dev) • All: link GitHub to Claude team account for automated commits; PR review process confirmed • Min: placeholder data sent to Prat so frontend work begins immediately without waiting for final data • Set up AWS environment: cloud database (PostgreSQL), file storage (S3), access permissions — free database schema tool to eliminate extra costs • Design database schema: 8 tables covering fields, wells, operators, production, reservoir parameters, cost data, CEOR benchmarks, operator engagement scores • Nalin + Zain: review existing Excel dataset (June 3–4) — understand structure, spot obvious gaps, note which regions are most complete • Communicate gaps to research team (June 5–6) — flag missing or suspicious fields, confirm what they can realistically fill by Month 2
June 8–14 Week 2	• Build data import pipeline: scripts that read the Excel datasets and load them into the database; parser must handle formatting changes as Excel sheets are updated • Build validation rules for each field: water cut 0–100%, API gravity positive, temperature in plausible range, etc. • Anomaly flagging: bad records get logged and flagged, not silently dropped • Source P&A cost estimates (June 8–10): Tigbourne has no P&A figures for most fields. Nalin and Zain to research industry benchmarks (onshore vs. offshore, by region) from SPE papers, SEC plugging cost disclosures, and operator annual reports — essential for NPV calculation • Min: deliver field-mapping doc (SPE paper fields mapped to database columns) • Finalise and lock database schema after seeing real data shape
June 15–21 Week 3	• Build core API endpoints (FastAPI): field list with filters, field detail, well detail, operator profile • Add filtering, sorting, and pagination — Prat needs this to build the data grid on the frontend • Publish API documentation (auto-generated) — this is the contract Prat builds the frontend against from this point forward • Zain: connect commodity price feed (Brent + WTI minimum; Dubai if available) — GET /prices/forward-curve endpoint • Prat: frontend connected to real API data — no more placeholder content for core screens
June 22–30 Week 4 — MVP	• Data export working: users can filter fields and export results as CSV, PDF, or DOCX • Optional: data visualisation graphs based on filtered results — add if time allows • Secure login system: tokens for authentication, user permission levels (admin, institutional, individual) • End-to-end test: filter data → view results → export — works without errors • MVP internal demo: team runs through the full product together before showing Buddy

Month 2 — July 2026: CEOR & Financial Engine

July 1–14 Weeks 5–6	• Break-even calculation engine: for any field, calculate the oil price at which it stops being profitable — using lifting + transport + marketing costs vs. live oil price • NPV engine: how much is a field worth today — discounting future cash flows and subtracting P&A costs • Scenario API endpoint: takes a field ID and assumptions (oil price, water cut, chemical type), returns CEOR vs. drilling comparison • Zain: help build and test break-even and NPV logic
July 15–28 Weeks 7–8	• Water cut intervention curve: calculate NPV at each intervention point — 10%, 20%... 90% — showing the value lost by waiting • CEOR technology benchmarks loaded: costs for polymer, surfactant, ASP, foam, and wettability alteration from supplier tech data • Validate all outputs against source Excel model for 3 reference fields — any discrepancy resolved before Month 3 • Min's SPE data pipeline integrated — parsed data flows into the database without manual reformatting • Zain: lead validation testing across all 3 reference fields

Month 3 — August 2026: Polish & Demo

Aug 1–21 Weeks 9–11	• Full frontend/backend integration — all screens connected to live data, no placeholder content remaining • End-to-end testing of every user flow: login, search, filter, CEOR analysis, export • Performance: database indexes on most-queried columns; basic caching on price feed • Website polished — must look professional to support Buddy's sales team and customer acquisition • Load at least 2 full regions (e.g. Azerbaijan + one Middle East region) with clean, validated data • API key delivery system: external subscribers receive API keys for programmatic access to formatted data
Aug 22–31 POC MILESTONE	• Demo environment set up — stable, separate from dev, seeded with clean data • Run full 30-minute demo script at least twice before UBS / Capital One meeting • All code documented — README updated; any team member can work independently without asking Nalin • POC sign-off checklist completed (see Section 7)

7. Success Criteria

MVP Sign-Off — End of June 2026

- **Export works:** Users can filter oil field data and export results in CSV, PDF, and DOCX — all three formats, no errors
- **Frontend connected:** Prat's interface runs on real data, not placeholder — Min's final data loaded and displaying correctly
- **Login works:** Users can sign up, log in, and see only the data their permission level allows
- **Clean and fast:** No crashes, no slow load times on the core filter and export flow

POC Sign-Off — Aug 31, 2026

- **Economics validated:** CEOR vs. drilling outputs match the source Excel model for all 3 reference fields — no unexplained differences
- **Water cut engine works:** Platform shows the value difference between intervening at 10% vs. 80% water cut for any field in the database
- **Live prices active:** Brent and WTI forward curves update automatically — financial calculations use live data
- **P&A included:** Net present value calculations include P&A cost estimates for all fields — no incomplete valuations
- **Demo-stable:** 30-minute demo to UBS or Capital One completes without crashes, missing data, or manual workarounds
- **Website looks professional:** Design is polished enough that Buddy's sales team can use it in outreach and customer acquisition
- **Code is documented:** README and Claude.md are current; any team member can work independently without asking Nalin

Long-Term (Months 3–6)

- **Scale-ready:** API handles multiple enterprise users simultaneously without slowing down
- **Operator ratings live:** Tigbourne's 3 years of engagement history modelled as a searchable score for each operator
- **Not worth building in-house:** Product is advanced enough that SLB, Baker Hughes, or Halliburton would rather subscribe — the acquisition-defence threshold

8. Infrastructure & Costs

Item	Estimated Cost	Who Pays	Notes
AWS server (cloud hosting)	\$50–100 / month	Buddy	Depends on website traffic volume
Domain + hosting	~\$200 one-time	Buddy	Covered by Buddy

Item	Estimated Cost	Who Pays	Notes
Database	\$0	—	Free schema management tool — no additional database costs
Claude API backup	Available	—	Backup account available if team runs out of tokens

- **Action:** Nalin and Zain to submit a full cost estimate to Buddy before finalising any infrastructure decisions.

9. Version Control & Code Standards

- **GitHub repo:** Created by Prat on June 1; all contributors (Nalin, Zain, Min) added immediately
- **Claude integration:** All GitHub accounts linked to the Claude team account — Claude can commit changes directly once linked
- **Pull request review:** Someone must review every pull request before merging to ensure changes do not break existing code
- **README:** Claude writes and updates README notes documenting all changes — team continuity depends on this
- **Claude.md:** Must include proper test design rules to prevent fake or meaningless tests — agreed in team meeting

10. Key Risks

Risk	Mitigation
Source data has known errors (mixed formats, zero values, stale entries)	Validate every record before loading; flag bad data visibly; return to source for correction. Data parser must handle ongoing Excel format changes.
P&A costs missing for most fields	Nalin and Zain research industry benchmarks in Week 2; use estimates flagged as such until real figures are available or (Delegate to research team)
Financial model translation errors	Validate against 3 reference fields before POC demo; any discrepancy blocks sign-off
Commodity price API gaps — Dubai may not be on free tier	Zain identifies primary and backup sources by June 7; Brent + WTI are the minimum or (Delegate to research team)
Scope creep — CEOR features tempt team to skip MVP basics	MVP scope (filter + export) must be complete and tested before any CEOR work begins
Integration failures between API and frontend	API documentation published June 21; not changed without notice; versioned from day one

11. Confidentiality

All content in this document is covered by the NDA and non-circumvention agreement signed with Tigbourne Capital. The platform concept, financial model, field data, operator engagement scores, and chemical supplier relationships are strictly confidential. Nothing may be shared outside the immediate project team without explicit written authorisation from Tigbourne Capital.

